ZACHARY YANINEK

Cambridge, MA

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EDUCATION

MASSACHUSETTS INSTITUTE OF TECHNOLOGY, SLOAN SCHOOL OF MANAGEMENT Cambridge, MA

Candidate for Master of Business Analytics, Operations Research Center, August 2027 2026 - Present

- Coursework: Machine Learning, Optimization Methods, Advanced Analytics Edge, Analytics Lab, AI Studio
- Machine Learning Project (MIT Sundai Hack): Built and deployed glucose app; XGBoost R^2 rose from 0.21 to 0.31 for unseen participants by adding fasting labs to meal log and pre-meal glucose (1,382 meals; FastAPI, React, Docker)
- AI Agent Project (MIT Media Lab AI Studio): Designed NANDA Town multi-agent fault test; all messages from 1 of 4 agents arrived garbled, yet both validators passed, checking what was sent, not what arrived; opened PR proposing fix
- Clubs: MIT Sloan Healthcare Club, MIT Figure Skating Club, Sundai Club

PRINCETON UNIVERSITY Princeton, NJ

B.S.E. in Operations Research and Financial Engineering, summa cum laude 2022 - 2026

- Honors: Phi Beta Kappa, Tau Beta Pi, Sigma Xi, Kenneth H. Condit Prize, Fulbright France grant (declined)
- Grant: Princeton ORFE Summer Research Grant (Vijay Kumar Gupta Fund) funded Institut Pasteur internship (2024)
- Teaching: Undergraduate Course Assistant for calculus and linear algebra (2024 - 2026)
- Minor: French

TECHNICAL SKILLS

- Machine Learning & Optimization: Python (PyTorch, XGBoost, scikit-learn, Optuna), Gurobi, PyEPO, CVXPY
- Deployment & AI Tools: FastAPI, React, Docker, SQL (SQLite), LLM agents (MCP), Claude Code, Codex

EXPERIENCE

TELEOS LABS Cambridge, MA

AI startup giving families one record of aging parent's care across home, hospital, and assisted living 2026 - Present

Co-founder & CEO *(Part-time)*

- Founded startup; wrote go/no-go test plan with kill criteria before first interview; built pitch decks and met investors
- Scripted and led customer interviews with family caregivers and care staff; families rejected apps that assign them tasks or make them log care, so product builds updates from records caregivers already keep
- Directed AI coding agents (Claude Code, Codex) to build and deploy synthetic-data prototype from written specs, reviewing every change (Python, SQLite, MCP, Docker)

PRINCETON UNIVERSITY Princeton, NJ

Machine learning and optimization for scheduling stroke patients' CT scans in neuro ICU 2025 - 2026

Senior Thesis Researcher *(Advisor: Prof. Bartolomeo Stellato; in collaboration with Prof. Agni Orfanoudaki, Oxford)*

- Lowered decision regret 35% vs MSE-trained network and captured 79.3% of oracle reward vs 68-71% for other ML models by training PyTorch network end-to-end through Gurobi optimizer (decision-focused learning, PyEPO)
- Simulated 30-patient neuro ICU over 1,000 hours, replaying held-out patients from 674-patient dataset, to evaluate scan policies under scanner breakdown and data drift; tuned models with 60-trial Optuna searches
- Adjusted for clinician selection bias in training targets with inverse propensity weighting (scikit-learn)

INSTITUT PASTEUR Paris, France

French nonprofit biomedical research foundation Summer 2024

Summer Research Intern *(Advisor: Prof. Marc Lecuit, MD)*

- Investigated *Listeria monocytogenes* virulence genes through bench work
- Modeled host infection progression and analyzed data in Python

ADDITIONAL INFORMATION

- Languages: French (fluent)
- Personal Project: Developed portfolio manager suggesting trades for own brokerage account (CVXPY); adding published cost-aware trading rule raised annualized return 6.9% to 8.6%, cut turnover 12.2x to 4.8x/year (2014 - 2026 backtest)
- Interests: Figure skater on U.S. Figure Skating Association National Development Team (2019 - 2022)